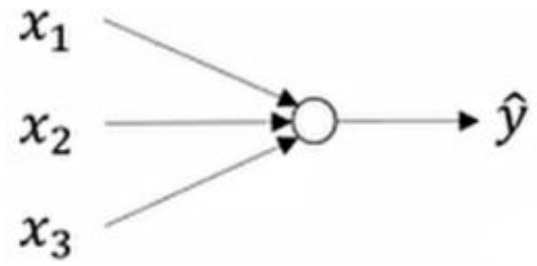
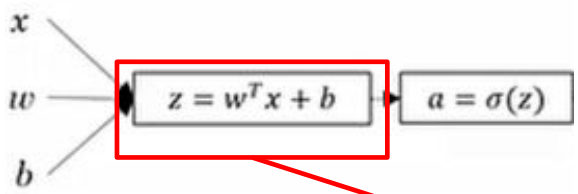
Neural Network Overview

미래연구소 16기 3주차

0. Week 2에서 배운 내용

		계산 과정
모델		(x = (x1, x2, x3), y)가 m개 주어진다. = (X, Y) 1) w1, w2, w3, b initalize 1) np.dot(w.T, X) + b => Z sigmoid(Z) => A = Y^ 1) compute J(w, b) 1) backpropagation => dw, db 1) w = w - αdw, b = b - αdb
computation graph		
	$np.dot(w.T, X) + b = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \\ x_3^{(1)} & x_3^{(2)} \end{bmatrix} + b$	

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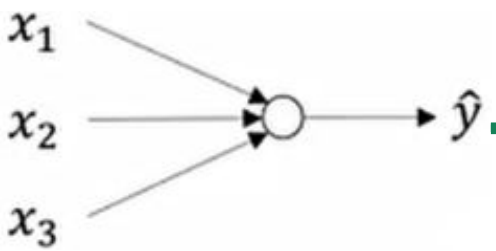
<새로운 표현법>

w 앞의 숫자: unit index = 현재 layer index

w 뒤의 숫자: feature index = 이전 layer index

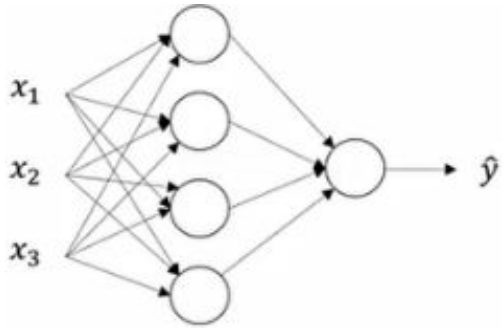
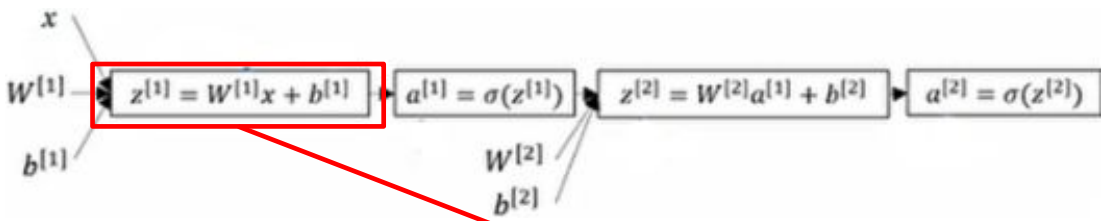


$$np.dot(w.T, X) + b = \begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1{}^{(1)} & x_1{}^{(2)} \\ x_2{}^{(1)} & x_2{}^{(2)} \\ x_3{}^{(1)} & x_3{}^{(2)} \end{bmatrix} + b$$

$$np.dot(w.T, X) + b = \begin{bmatrix} -w.T- \end{bmatrix} \begin{bmatrix} x_1{}^{(1)} & x_1{}^{(2)} \\ x_2{}^{(1)} & x_2{}^{(2)} \\ x_3{}^{(1)} & x_3{}^{(2)} \end{bmatrix} + b$$

$$np.dot(w.T, X) + b = \begin{bmatrix} w_{1\ 1} & w_{1\ 2} & w_{1\ 3} \end{bmatrix} \begin{bmatrix} x_1{}^{(1)} & x_1{}^{(2)} \\ x_2{}^{(1)} & x_2{}^{(2)} \\ x_3{}^{(1)} & x_3{}^{(2)} \end{bmatrix} + b$$

1. Week 3에서 배울 내용

모델		week 2 모델 연산에서 forward propagation backward propagation 부분만 2배로 증가
computation on graph		
	$np.dot(W^{[1]}, X) + b = \begin{bmatrix} w_{1\,1} & w_{1\,2} & w_{1\,3} \\ w_{2\,1} & w_{2\,2} & w_{2\,3} \\ w_{3\,1} & w_{3\,2} & w_{3\,3} \\ w_{4\,1} & w_{4\,2} & w_{4\,3} \end{bmatrix} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \\ x_3^{(1)} & x_3^{(2)} \end{bmatrix} + b$	

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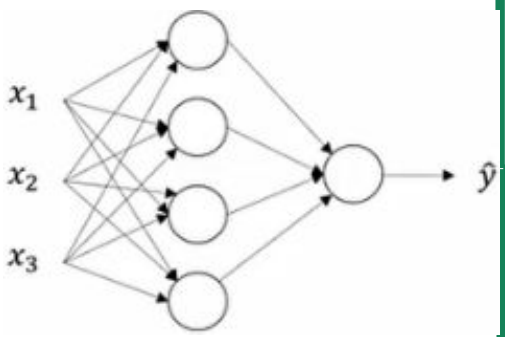

$$np.dot(w.T, X) + b = \begin{bmatrix} w_{1\ 1} & w_{1\ 2} & w_{1\ 3} \\ w_{2\ 1} & w_{2\ 2} & w_{2\ 3} \\ w_{3\ 1} & w_{3\ 2} & w_{3\ 3} \\ w_{4\ 1} & w_{4\ 2} & w_{4\ 3} \end{bmatrix} \begin{bmatrix} x_{1}^{(1)} & x_{1}^{(2)} \\ x_{2}^{(1)} & x_{2}^{(2)} \\ x_{3}^{(1)} & x_{3}^{(2)} \end{bmatrix} + b$$

$$np.dot(W^{[1]}, X) + b = \begin{bmatrix} -w_1\ T- \\ -w_2\ T- \\ -w_3\ T- \\ -w_4\ T- \end{bmatrix} \begin{bmatrix} x_{1}^{(1)} & x_{1}^{(2)} \\ x_{2}^{(1)} & x_{2}^{(2)} \\ x_{3}^{(1)} & x_{3}^{(2)} \end{bmatrix} + b$$

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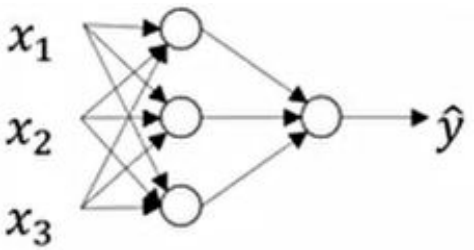

$$np.dot(w.T, X) + b = \begin{bmatrix} w_{1\ 1} & w_{1\ 2} & w_{1\ 3} \\ w_{2\ 1} & & w_{2\ 3} \\ w_{3\ 1} & w_{3\ 2} & w_{3\ 3} \\ w_{4\ 1} & w_{4\ 2} & w_{4\ 3} \end{bmatrix} \begin{bmatrix} x_1^{(1)} & x_1^{(2)} \\ x_2^{(1)} & x_2^{(2)} \\ x_3^{(2)} & \end{bmatrix} + b$$

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2. Week 3에서 주되게 다뤄질 내용

1) hidden layer의 등장: lec 2

1) hidden unit(node)가 여러 개인 상황: lec 3 ~ lec 5

1) activation function: lec 6 ~ 8

1) hidden layer가 있을 때의 5가지 step 진행: lec 9 ~ lec 10

1) initialization (unit이 여러 개인 상황): lec 11